



FIRST SEMESTER, 2026–2027
Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : ME F427
Course Title : Continuum Mechanics
Instructor-in-Charge : Prof. Amit R. Singh

1. Course Description:

Introduction to tensors, tensor algebra and tensor calculus. Kinematics of deformation (displacement, velocity, deformation gradient, material and spatial field descriptions, description of local deformation, linearized kinematics, kinematic rates), left & right Cauchy-Green deformation tensor, Cauchy-Lagrangian strain tensor. Traction and stress (first Piola-Kirchhoff stress, second Piola-Kirchhoff stress, Biot stress etc). Mechanical conservation and balance laws (conservation of mass, conservation of linear momentum, conservation of angular momentum, conservation of energy, work done by stresses and the principle of virtual work, BVP, IVP, calculus of variation). Constitutive relations (constraints on constitutive relations, frame indifference, objectivity, material symmetry, constitutive models for linearly elastic, hyperelastic solids and fluid flows).

2. Scope & Objective of the Course:

This course will provide an introduction to continuum mechanics. It aims to develop in students the foundational mathematical rigor needed to understand advanced topics in solid and fluid mechanics.

3. Textbooks:

TB1. *Continuum Mechanics for Engineers*, G. T. Mase, R. E. Smelser, J. S. Rossmann, 4th edition, CRC Press, 2010

References:

- Ref1.** *Introduction to Continuum Mechanics*, W. Michael Lai, David Rubin, and Erhard Krepl, 4th edition, Butterworth-Heinemann, 2009
- Ref2.** *Continuum Mechanics, Concise Theory and Problems*, Peter Chadwick, 2nd edition, Dover Publications, 1999
- Ref3.** *Principles of Continuum Mechanics*, J. N. Reddy, 2nd edition, Cambridge University Press, 2018

4. Course Plan:

Module	Lectures	Reference	Learning Outcomes
Tensors	1-4	T1 Ch 2	To learn indicial notation, algebra and calculus of tensors
Kinematics	5-13	T1 Ch 4	To learn description of deformation and related quantities
Stress	14-21	T1 Ch 3	To learn various measures of stress
Conservation Laws	22-30	T1 Ch 5	To derive fundamental laws of mechanics
Constitutive Relations	31-40	T1 Ch 5, 6	To explore various stress-strain relationships

5. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Quiz	30	15 mins.	To be announced	Closed Book, best 3 of 5
Mid-Semester test	30	90 mins.	7/10 FN1	Closed Book
Comprehensive Examination	40	180 mins.	8/12 FN	1 A4-size sheet allowed

NC Criterion: Skipping all the lab assignments or both the quizzes or the mid-semester exam or the final exam will result in a NC grade.

6. Office Consultation Hour:

Thursday 4 pm to 4:50 pm. We can also set up a time by appointment via email.

7. Make-up Policy:

Make-up policy for the Mid-semester exam and the comprehensive exam will be as per AUGS guidelines.

Prof. Amit R. Singh
Instructor-in-charge
ME F427 Continuum Mechanics